



Saveetha Institute of Medical and Technical Sciences



A Secure LAN-Based File Sharing System with Authentication and Access Control

CSA0716 – Computer Networks

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INTRODUCTION

- A **Secure LAN-Based File Sharing System with Authentication and Access Control** is designed to enable users within a **Local Area Network (LAN)** to securely share files while protecting sensitive data from unauthorized access. Unlike traditional file-sharing methods, this system verifies the identity of users through a secure authentication process before granting access to shared resources.
- To enhance security, files are encrypted before transmission, ensuring that intercepted data cannot be read by unauthorized users. The system also implements **role-based access control**, allowing different levels of permissions such as administrator and standard user, thereby restricting access based on user roles.
- Additionally, all file-sharing activities are recorded in activity logs, providing accountability and improving system monitoring. This solution offers a secure, efficient, and reliable approach for file sharing in organizations, educational institutions, and office environments.





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Problem Statement

- Traditional file-sharing methods within a Local Area Network (LAN) often lack strong security mechanisms, making them vulnerable to unauthorized access, data breaches, and file tampering.
- Many systems do not verify user identity before granting access, allowing unauthorized users to view or modify sensitive files.
- Additionally, files transmitted without encryption can be intercepted by attackers on the network, compromising data confidentiality.
- The absence of proper access control and activity logging further makes it difficult to monitor user actions and detect security violations.
- Therefore, there is a need for a secure LAN-based file sharing system that integrates user authentication, file encryption, role-based access control, and activity logging to ensure safe and reliable file sharing within an organization.

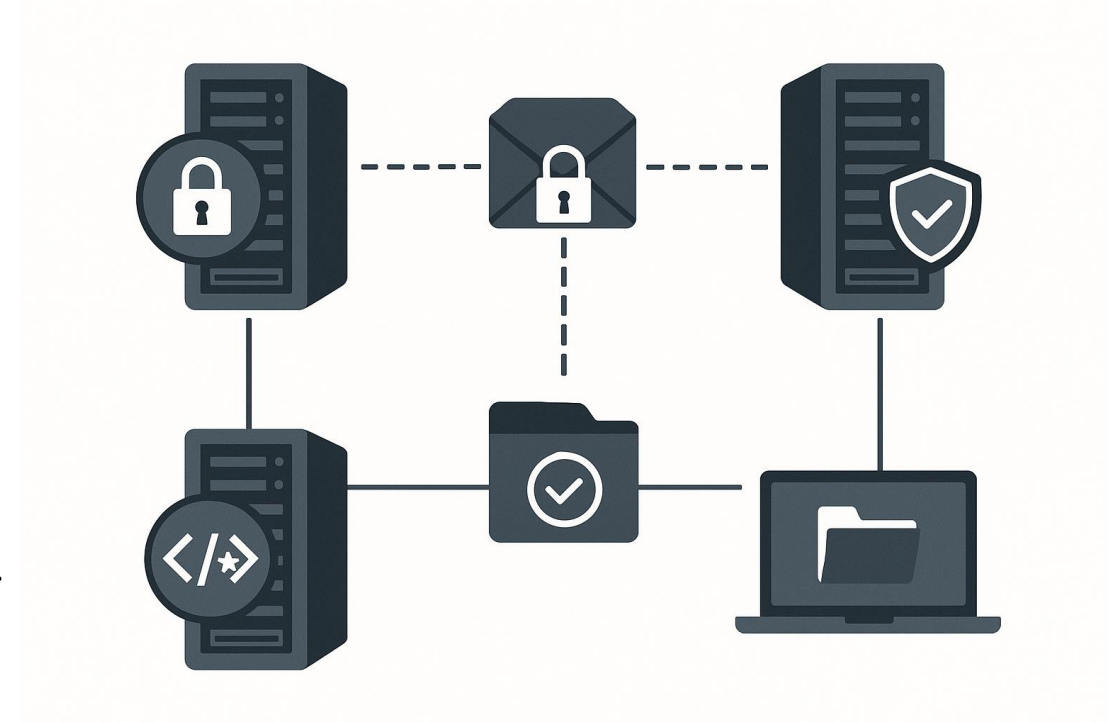


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Objectives

- To develop a secure file sharing system for users within a Local Area Network (LAN).
- To provide secure user login using authentication.
- To encrypt files before sharing to protect sensitive information.
- To allow only authorized users to access shared files.
- To enable secure file transfer between the client and server.
- To maintain records of file uploads, downloads, and user activities.
- To monitor network communication using Wireshark.
- To provide a safe, reliable, and easy-to-use file sharing solution.





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Software Required:

Software	Purpose
Python 3.x	Develop the client-server application and implement security features.
Visual Studio Code	Write, edit, and debug the Python code.
SQLite / MySQL	Store user accounts, passwords, and activity logs.
Python Socket Library	Enable communication between the client and server over the LAN.
Cryptography Library (AES)	Encrypt and decrypt files during transfer.
Wireshark	Monitor and analyze network packets to verify secure communication.
Git (Optional)	Manage source code versions and project updates.



Module 1: User Authentication

Description

This module verifies the identity of users before allowing access to the file sharing system. It provides a secure login process by validating user credentials stored in the database.

Functions

- User Registration
- User Login
- Password Verification
- User Authentication

Input

- Username
- Password

Output

- Authenticated user access to the system



Module 2: Secure File Transfer

Description

This module enables secure file transfer between the client and server over a Local Area Network (LAN). It establishes a connection using Python socket programming and transfers files only after successful user authentication.

Functions

- Establish Client-Server Connection
- Upload Files
- Download Files
- Secure File Transfer

Input

- Selected file
- Transfer request

Output

File successfully transferred between the client and server



Module 3: Encryption and Access Control

Description

This module encrypts files before transmission to protect sensitive data and ensures that only authorized users can access shared files based on their assigned permissions.

Functions

- AES File Encryption
- AES File Decryption
- Role-Based Access Control
- Permission Verification

Input

- File selected for sharing
- User access request

Output

Encrypted file with secure access granted only to authorized users



Module 4: Activity Logging and Network Monitoring

Description

This module records all user activities, such as file uploads, downloads, and login attempts, while monitoring network traffic using Wireshark to verify secure communication.

Functions

- Record User Activities
- Log File Uploads and Downloads
- Monitor Network Traffic
- Verify Secure Data Transmission

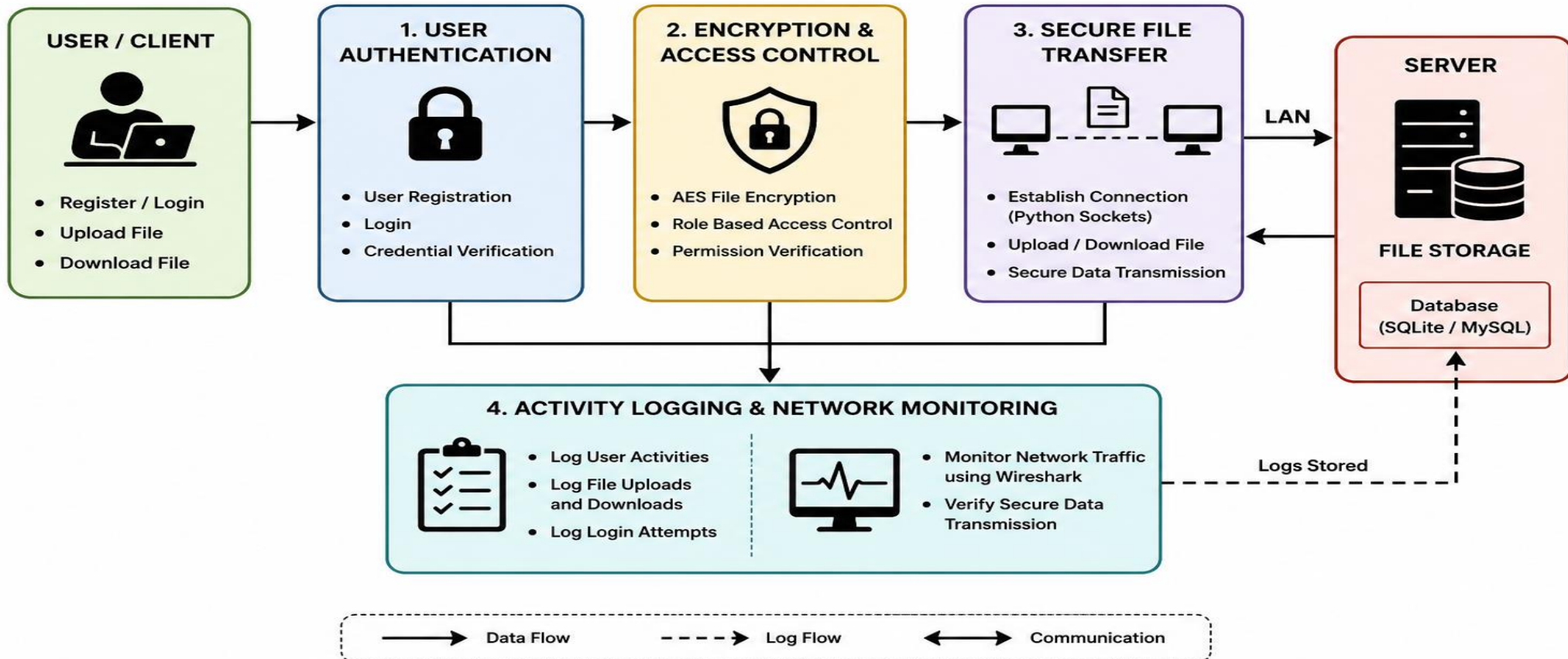
Input

- User actions
- Network packets

Output

- Activity logs
- Network monitoring report showing secure file transmission

BLOCK DIAGRAM SECURE LAN BASED FILE SHARING SYSTEM



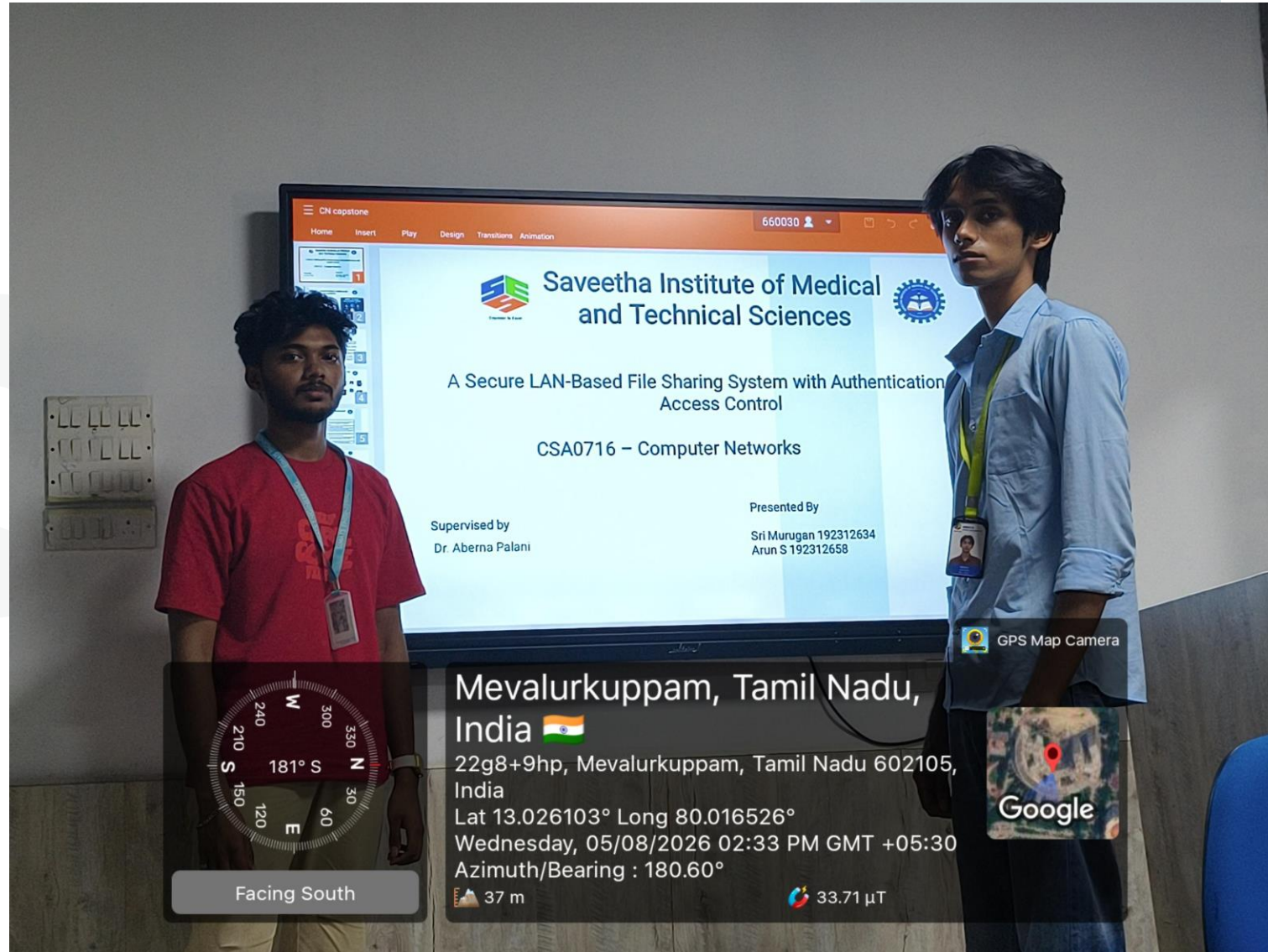


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Conclusion

The **Secure LAN-Based File Sharing System with Authentication and Access Control** provides a secure and efficient solution for sharing files within a Local Area Network. By integrating **user authentication, AES file encryption, secure client-server communication, role-based access control, and activity logging**, the system protects sensitive data from unauthorized access and ensures secure file transmission. It is a reliable solution for organizations, educational institutions, and office environments where secure and controlled file sharing is essential.



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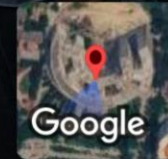
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THANK YOU